

MANAS SHARMA

B.Tech Data Science Undergraduate — AI & Machine Learning Researcher

Research Focus: Explainable Reinforcement Learning for Intelligent Systems

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IELTS: Overall 6.5 (All Bands 6.5)

RESEARCH INTERESTS

Reinforcement Learning, Explainable Artificial Intelligence (XAI), Machine Learning Systems, Intelligent Network Optimization, AI-Native Communication Systems

EDUCATION

B.Tech in Data Science

SRM Institute of Science and Technology, India

CGPA: 8.33 / 10

Expected Graduation: May 2026

RESEARCH EXPERIENCE

AI-Native Network Optimization Research

- Designed reinforcement learning-based framework for adaptive resource allocation in intelligent communication networks.
- Investigated optimization strategies for bandwidth prioritization using intent-driven automation.
- Conducted experimental evaluation of intelligent decision-making models for next-generation infrastructure systems.

Explainable Machine Learning Systems

- Developed interpretable machine learning models using SHAP-based feature attribution techniques.
- Studied transparency and trust mechanisms in AI-driven decision-making pipelines.

PUBLICATIONS

Sharma, M. "Reevaluating CNN Filter Dimensions: Experimental Findings on CIFAR-10 and Fashion-MNIST"
ICICV 2026 — Accepted for Oral Presentation

Sharma, M. "Intent-Driven AI-Native Network Slicing for Rural Broadcasting over ATSC 3.0/B2X"
WOCC 2026 — Under Review (EDAS)

RESEARCH & INDUSTRY EXPERIENCE

Machine Learning Intern 1Stop, Chennai

Jun 2024 – Jul 2024

- Developed supervised learning models (Random Forest, XGBoost) on datasets exceeding 10k records.
- Improved predictive performance through feature engineering and hyperparameter optimization.
- Evaluated models using cross-validation, precision, recall, and statistical performance metrics.

SELECTED RESEARCH PROJECTS

Intent-Driven ATSC Network Slicing (IIT Delhi Winner)

- Proposed reinforcement learning-based adaptive bandwidth allocation framework.
- Designed AI-native telecom slicing architecture for intelligent broadcasting systems.

6G Intelligent Network Management System

- Developed analytics-driven architecture enabling adaptive resource orchestration in next-generation networks.

Explainable AI Loan Prediction System

- Implemented interpretable machine learning framework using SHAP for transparent decision analysis.

AI Traffic Management System

- Developed OpenCV-based real-time traffic congestion detection and density estimation pipeline.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Scikit-learn, XGBoost, Ensemble Learning

Deep Learning: Neural Networks, Computer Vision, NLP, Reinforcement Learning

Data Science: Pandas, NumPy, Statistical Analysis, Visualization

Tools: Git, Docker, MongoDB, FastAPI, Linux

Concepts: Optimization Algorithms, Model Deployment, REST APIs

ACHIEVEMENTS

- 1st Prize — Build-a-thon 3.0, ITU IIT Delhi
- 1st Prize — Build-a-thon 4.0, ITU
- 2nd Prize — AI Agentic Battle

LANGUAGES

Hindi (Native), English (Professional), German (Beginner)